Computer Security

Riassunto completo del corso

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1 Prefazione

1.1 Disclaimer

I contenuti di questo PDF sono stati compilati con il solo scopo di avere una risorsa esaustiva e facilmente consultabile degli argomenti trattati nel corso di Computer Security sostenuto dal professore Zanero: non sono in alcun modo materiale sostitutivo alle lezioni, che frequentate in presenza mi hanno permesso di capire a fondo la materia. Sconsiglio di usare esclusivamente suddetto documento per cercare di prepararsi all'esame, supporto invece l'uso di questa dispensa per fare il punto della situazione durante il periodo di studio.

1.2 Keywords

2 Introduction

How can we define what is a secure system? Usually, it follows the **CIA** paradigm to correctly define what a secure system is, the problem lies inside its definition:

- **Confidentiality** - the system should not leak information to unauthorized parties.
- **Integrity** - the system should not allow unauthorized parties to modify data **and** should not allow authorized parties to modify data in an unauthorized way.
- **Availability** - information should be always available to authorized parties.

The problem lies inside **Integrity**, if deciding whether a party is authorized or not is simple since it's a binary decision, the decision of whether a modification is authorized or not is much more complex and involves a lot of factors. Also, **Availability** is what pays off in the end. What can we consider **secure**? Something is secured if it takes an **effort** to break it: the more effort it takes, the more secure it is.

Secure System

A **Secure System** is a system that requires a lot of effort to be broken.

2.1 Key concepts

We firslty need some definitions useful to understand the world of security:

- **Vulnerability** - It is a shortcut that allows an attacker to break the system with less of the intended effort: it is something that allows to violate one of the CIA properties.
- **Exploit** - It is a specific way to use one or more vulnerabilities to accomplish a specific objective that violates one of the CIA properties.
- **Asset**
- **Threat**
- **Risk**

Understanding what is and what isn't a vulnerability

Not all vulnerabilities lookalikes are actual vulnerabilities: sometimes, something looks vulnerable but the accomplishment of the attack doesn't result in a violation of the CIA properties.

2.1.1 Example: exploiting a vulnerability

When exploiting a vulnerability, we firstly need to find the security assumption that lays behind the system: in this example, the thing that ensures that the lock works is the fact that the pins are being pushed down always by the strings of the lock, so the only way to open the lock is to push up all the pins at the same time. Lockpicking is using a vulnerability of the lock to open it: this is using an **exploit**, using something to take advantage of a vulnerability to break the system. How can we resolve this vulnerability? Short answer: we make it harder to exploit

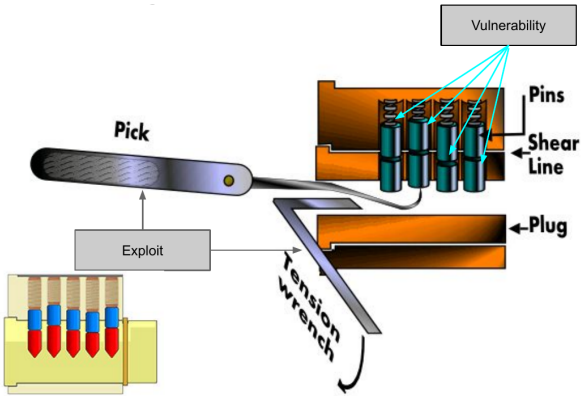

Figure 1: Lockpicking example

it. Oiled locks are harder to exploit because of the deletion of friction, but still are vulnerable: some vulnerabilities are implicit in the design of the system itself.